Sleep Apnea Detection from Physiological Signals

AI for Health — SRIP 2026 | Assignment Report

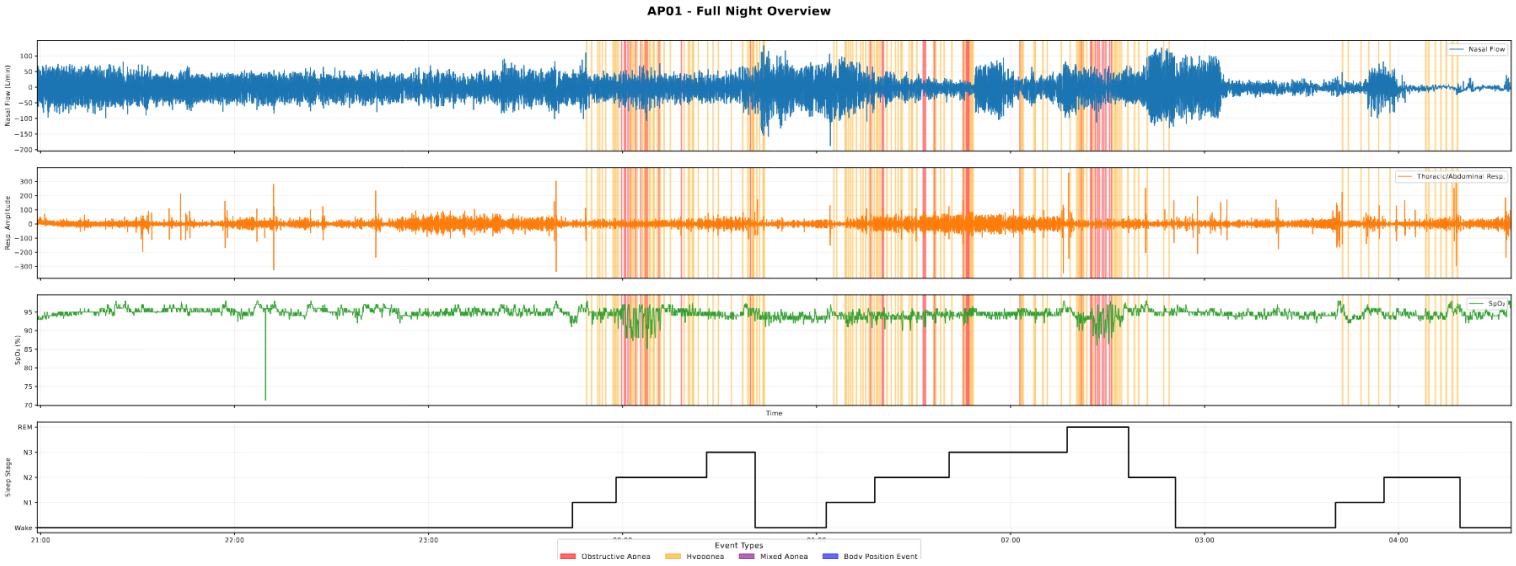
1. Introduction

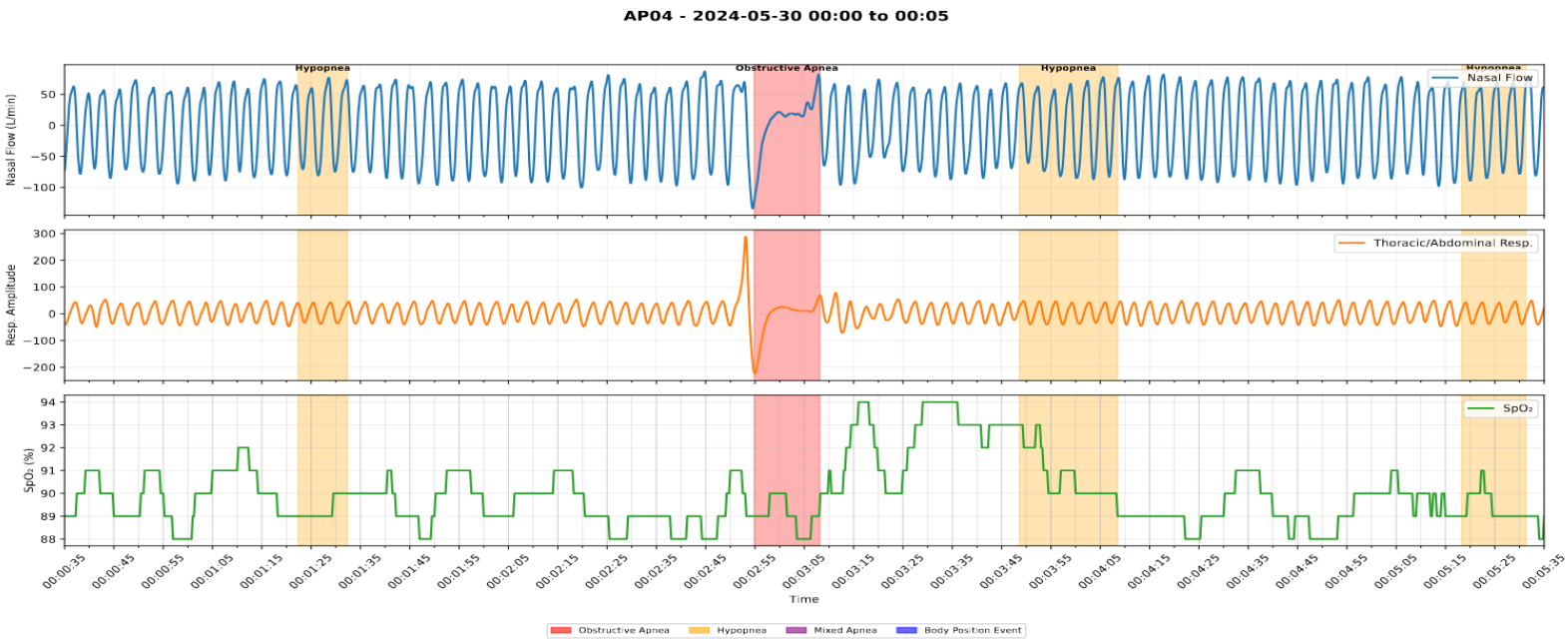
This project analyzes overnight physiological signals to detect abnormal breathing patterns during sleep. The dataset contains nasal airflow, thoracic movement, and SpO2 signals from five participants, each representing an approximately 8-hour nocturnal recording, along with annotated breathing events. The objective is to preprocess these signals, generate labeled time windows, and train a machine learning model to classify windows as **normal or abnormal breathing events** using Leave-One-Participant-Out (LOPO) cross-validation

2. Data Visualization

For each participant, nasal airflow (32 Hz), thoracic movement (32 Hz), and SpO2 (4 Hz) signals were plotted over the full 8-hour recording. Annotated breathing events were overlaid as color-coded highlights: Obstructive Apnea (red), Hypopnea (orange), Mixed Apnea (purple), and Body Position Events (blue). Signals were aligned using their timestamps despite different sampling rates.

The visualization script generates a PDF for each participant containing two components: a full-session overview showing the entire night on one page, and a series of 5-minute detail pages with finer time resolution. This allows both high-level pattern inspection and granular review of individual events.





3. Signal Preprocessing

3.1 Filtering

The nasal airflow signal was filtered using a 4th-order Butterworth bandpass filter retaining the breathing frequency range of 0.17-0.4 Hz, corresponding to normal human breathing rates of approximately 10-24 breaths per minute. The filter was applied using zero-phase filtering (filtfilt) to avoid phase distortion.

3.2 Windowing

The filtered nasal airflow signal was segmented into 30-second windows with 50% overlap (15-second step size). At 32 Hz, each window contains 960 samples. Overlapping windows increase the number of training samples and reduce boundary effects. The window start and end times were tracked using the original datetime index for accurate label assignment.

3.3 Labeling

Each window was assigned a binary label using the annotated event file. A window was labeled Abnormal (1) if any single annotated event overlapped with more than 50% of the window duration (greater than 15 seconds). Otherwise the window was labeled Normal (0).

3.4 Binary vs. Multi-Class Formulation

The raw annotations contain multiple event types: Hypopnea, Obstructive Apnea, Mixed Apnea, and Body Position Events. With only 5 participants, several categories appear extremely rarely — body events occur only a handful of times across the entire dataset. Multi-class classification under LOPO cross-validation compounds this further: a rare event type present in training participants may be completely absent in the held-out test participant, making metrics unstable and generalization unreliable. The task was therefore reformulated as binary classification, merging all abnormal event types into a single Abnormal class. This improves training stability and lets the model focus on detecting the presence of a breathing irregularity rather than distinguishing rare subtypes.

4. Modeling and Evaluation

4.1 Model Architecture

A simple 1D Convolutional Neural Network (CNN) was trained to classify 30-second signal windows as Normal or Abnormal. The architecture consists of two Conv1D blocks (16 and 32 filters) each followed by ReLU activation and max pooling, a global average pooling layer to collapse the temporal dimension, and two fully connected layers (32 units, dropout 0.5) leading to a single sigmoid output.

4.2 Training Setup

The model was trained using the Adam optimizer (lr=0.001) for 15 epochs with a batch size of 64. Class imbalance was handled via BCEWithLogitsLoss with pos_weight set to the ratio of normal to abnormal windows in each fold's training set, penalizing missed abnormal detections more heavily. Input windows were z-score normalized using mean and standard deviation computed from the training set only to prevent leakage.

4.3 Evaluation Strategy

Leave-One-Participant-Out (LOPO) cross-validation was used: in each of 5 folds, data from 4 participants was used for training and the remaining participant served as the held-out test set. This evaluates genuine cross-subject generalization and prevents subject-level data leakage that would occur with random splits.

5. Results

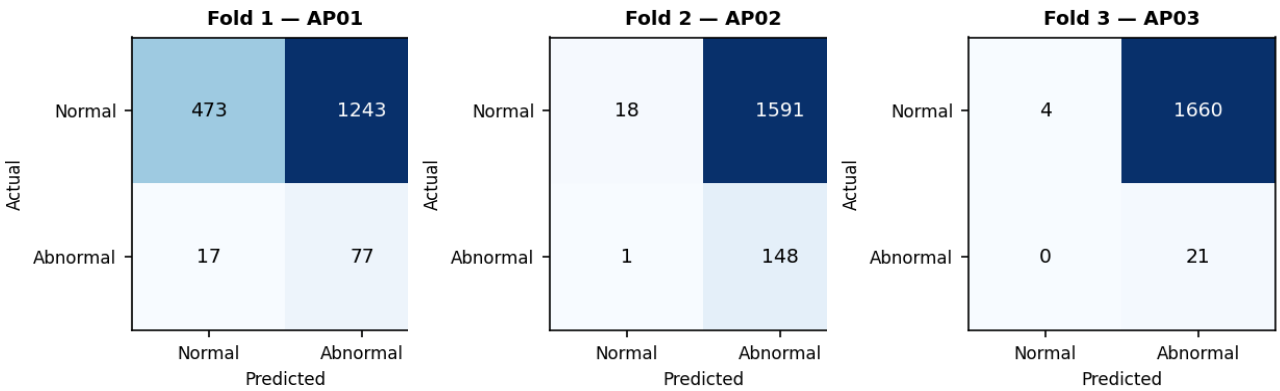
5.1 Per-Fold Metrics

Fold	Test Participant	Accuracy	Precision	Recall	F1
1	AP01	0.30	0.06	0.82	0.11
2	AP02	0.09	0.09	0.99	0.16
3	AP03	0.01	0.01	1.00	0.02
4	AP04	0.51	0.11	0.62	0.19
5	AP05	0.41	0.18	0.51	0.26
	Average	**0.26**	**0.09**	**0.79**	**0.15**

Table 1. Per-fold LOPO results. Precision, Recall, and F1 reported for the Abnormal class.

5.2 Confusion Matrices

The confusion matrices below show prediction counts for each fold. The model consistently predicts Abnormal for the majority of windows, resulting in high recall but low precision across all folds.



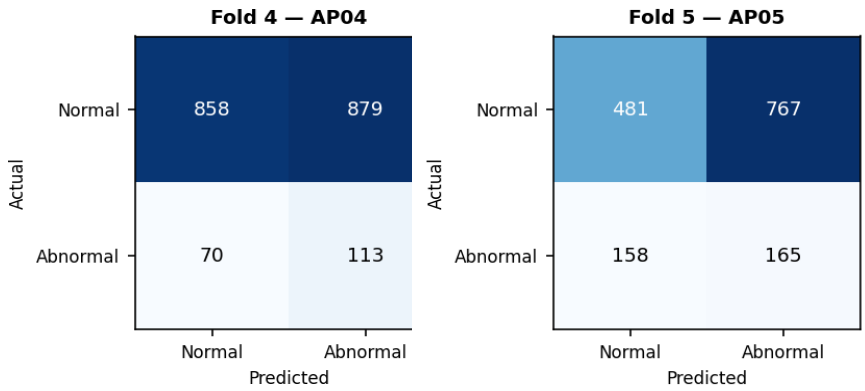


Figure 1. Confusion matrices for all 5 LOPO folds.

6. Additional Experiments — Baseline Models

To contextualize CNN performance, Logistic Regression and XGBoost baselines were evaluated using the same LOPO setup on the same windowed dataset.

Model	Accuracy	Precision	Recall	F1
Logistic Regression	~0.63	~0.16	~0.68	~0.26
XGBoost	~0.79	~0.16	~0.34	~0.21
1D CNN (ours)	~0.26	~0.09	~0.79	~0.15

Table 2. Comparison of baseline models under LOPO cross-validation.

XGBoost achieved the highest accuracy (0.79) but low recall (0.34), meaning it largely predicts Normal and misses most abnormal events. The CNN trades accuracy for recall — it detects more abnormal windows but at the cost of many false positives. In clinical screening contexts, high recall is generally preferable to high accuracy when the cost of missing an event is high.

7. Discussion

The CNN achieved high recall (0.79) but very low precision (0.09), meaning it detects most abnormal breathing events but produces many false positives. This behavior is primarily driven by strong class imbalance — normal windows significantly outnumber abnormal ones — and by the small dataset size of five participants.

Performance varies considerably across folds (accuracy ranging from 0.01 to 0.51), indicating subject-level distribution shift. Abnormality prevalence differs substantially across participants, making it difficult for the model to learn a consistent decision boundary. This is exactly the type of challenge LOPO evaluation is designed to expose and is a known difficulty in physiological signal modeling on small cohorts.

Possible improvements include incorporating additional signal channels (thoracic movement and SpO2) as a multi-channel input, applying stronger data augmentation or oversampling strategies, using early stopping to prevent overfitting, and evaluating on larger participant cohorts. Feature-engineered inputs (statistical summaries, frequency-domain features) rather than raw windows could also reduce sensitivity to signal-level variation across subjects.

8. Use of AI Tools

AI-assisted tools were used during development for clarification of technical concepts, debugging support, and assistance with code formatting and documentation. These tools were also helpful for reviewing parts of the codebase and identifying potential issues during development.

All core implementation, preprocessing design, experimental setup, and evaluation decisions were carried out independently by me.